LIBRARY MANAGEMENT SYSTEM

Deliverable 1 Document

Course: Software Engineering
Team Name: Stack Architects
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1. Introduction

The Library Management System (LMS) is a Java-based business management system designed to automate and streamline the operations of a library. Traditional manual library systems are inefficient, time-consuming, and prone to human errors. The proposed system aims to digitize book catalog management, user management, borrowing workflows, fine handling, and reporting activities.

The system will be developed using Java and object-oriented design principles. Scrum agile methodology is adopted to incrementally develop the system across multiple sprints.

2. Project Vision

The vision of the Library Management System is to create a secure, efficient, and user-friendly platform that automates all major library operations. The system aims to:

- Replace manual record keeping with a centralized digital system.
- Improve efficiency in issuing and returning books.
- Provide real-time availability tracking of books.
- Automate fine calculation and payments.
- Provide transparency and monitoring through administrative reports.
- Ensure scalability for future feature enhancements.

The long-term goal is to build a reliable system that improves operational efficiency while maintaining proper role-based access control.

3. Intended Use of the System

3.1 Stakeholders

The primary stakeholders of the system are:

- Students (Library Members)
- Librarians
- System Administrator
- University/Institution Management

3.2 Stakeholder Needs

Students

- Create an account (subject to admin approval)
- Search and view available books
- Request books for borrowing
- Reissue books (within allowed limits)
- View borrowing history
- Pay fines online
- View fine status

Librarians

- Manage book catalogue
- Approve or reject borrowing requests
- Record book returns
- Monitor overdue books
- Generate and monitor fines
- Contact students regarding overdue books

Administrator

- Approve account creation and deletion
- Manage user accounts
- Monitor fine collection
- Generate reports
- View system usage statistics

Management

- Monitor library performance

- View reports on borrowing activity
- Analyze fine collection statistics

4. Features / Overall Functionality

The Library Management System consists of the following modules:

4.1 Account Management

- Student registration (pending admin approval)
- Account deletion request
- Admin approval/rejection of accounts
- Role-based login system

4.2 Book Catalogue Management

- Add new books
- Update book details
- Delete books
- Maintain book quantity
- Categorize books

4.3 Search and Viewing

- Search books by title
- Search books by author
- Search books by category
- Search books by ISBN
- View availability status

4.4 Borrowing Workflow

- Student requests book
- Librarian approves/rejects request
- System records issue date and due date
- Book quantity automatically updated

4.5 Reissue Management

- Student can reissue book within allowed period

- Reissue allowed only if:
    - Book not reserved by another user
    - No unpaid fine
    - Maximum reissue limit not exceeded

4.6 Return and Fine Management

- Librarian records book return
- System automatically calculates overdue fine
- Student can pay fine online
- Librarian can verify payment status

4.7 Reservation System

- Student can reserve unavailable books
- Librarian can manage or cancel expired reservations

4.8 Reporting and Monitoring

- Issued books report
- Overdue books report
- Fine collection report
- User activity report

5. User Stories

US-01 – Student Account Creation Request

Story ID: US-01
Story Title: Student Account Creation Request
Importance: High
Estimate: 3 hours

User Story:
As a: Student
I want: to request account creation
So that: I can access the library system and use its services

Acceptance Criteria:
And I know I am done when: A registration request is successfully submitted and appears in the

Admin's pending approvals list. The student receives a confirmation that their request is under review.

US-02 – Approve/Reject Account Creation

Story ID: US-02
Story Title: Approve or Reject Account Creation
Importance: High
Estimate: 2 hours

User Story:
As an: Admin
I want: to approve or reject account creation requests
So that: only valid users gain access to the system

Acceptance Criteria:
And I know I am done when: The admin can view pending requests, approve or reject them, and the student is notified of the decision.

US-03 – Account Deletion Request

Story ID: US-03
Story Title: Account Deletion Request
Importance: Medium
Estimate: 2 hours

User Story:
As a: User
I want: to request account deletion
So that: my account can be securely removed

Acceptance Criteria:
And I know I am done when: The deletion request is submitted and visible to the Admin for approval.

US-04 – Approve/Reject Account Deletion

Story ID: US-04
Story Title: Approve or Reject Account Deletion
Importance: Medium
Estimate: 2 hours

User Story:
As an: Admin
I want: to approve or reject deletion requests
So that: system data remains controlled and secure

Acceptance Criteria:
And I know I am done when: The admin approves/rejects the request and the user is notified. Approved accounts are removed or deactivated.

Book Management

US-05 – Add New Book

Story ID: US-05
Story Title: Add New Book
Importance: High
Estimate: 3 hours

User Story:
As a: Librarian
I want: to add new books
So that: they become available in the system

Acceptance Criteria:
And I know I am done when: The book details are saved and visible in the catalogue.

US-06 – Update Book Details

Story ID: US-06
Story Title: Update Book Information
Importance: High
Estimate: 2 hours

User Story:
As a: Librarian
I want: to update book details
So that: information remains accurate

Acceptance Criteria:
And I know I am done when: Updated information is reflected in the catalogue immediately.

US-07 – Delete Book

Story ID: US-07
Story Title: Delete Book
Importance: Medium
Estimate: 2 hours

User Story:
As a: Librarian
I want: to delete books
So that: outdated records are removed

Acceptance Criteria:
And I know I am done when: The selected book is removed from the system and is no longer searchable.

US-08 – Maintain Book Quantity

Story ID: US-08
Story Title: Maintain Book Quantity
Importance: High
Estimate: 2 hours

User Story:
As a: Librarian
I want: to maintain book quantity
So that: stock levels remain accurate

Acceptance Criteria:
And I know I am done when: Quantity updates correctly when books are issued or returned.

Search & Viewing

US-09 – Search Books

Story ID: US-09
Story Title: Search Books

Importance: High
Estimate: 3 hours

User Story:
As a: Student
I want: to search books
So that: I can find specific titles or authors

Acceptance Criteria:
And I know I am done when: Search results are displayed accurately based on title, author, or category.

US-10 – View Book Details

Story ID: US-10
Story Title: View Book Details
Importance: High
Estimate: 2 hours

User Story:
As a: Student
I want: to view book details
So that: I can check availability and borrowing status

Acceptance Criteria:
And I know I am done when: Book information such as title, author, availability, and quantity is displayed clearly.

Borrowing Workflow

US-11 – Request Book Borrowing

Story ID: US-11
Story Title: Request Book
Importance: High
Estimate: 3 hours

User Story:
As a: Student
I want: to request a book
So that: I can borrow it

Acceptance Criteria:
And I know I am done when: The borrowing request is sent to the Librarian and appears in the pending request list.

US-12 – Approve/Reject Borrowing Request

Story ID: US-12
Story Title: Approve or Reject Borrowing
Importance: High
Estimate: 3 hours

User Story:
As a: Librarian
I want: to approve or reject borrowing requests
So that: borrowing is properly controlled

Acceptance Criteria:
And I know I am done when: The request status updates to approved/rejected and the student is notified.

Reissue

US-13 – Reissue Book

Story ID: US-13
Story Title: Reissue Borrowed Book
Importance: Medium
Estimate: 2 hours

User Story:
As a: Student
I want: to reissue a borrowed book
So that: I can extend my borrowing period

Acceptance Criteria:
And I know I am done when: The due date is extended automatically within allowed limits without librarian approval.

Return & Fine Management

US-14 – Record Book Return

Story ID: US-14
Story Title: Record Book Return
Importance: High
Estimate: 2 hours

User Story:
As a: Librarian
I want: to record book return
So that: availability is updated

Acceptance Criteria:
And I know I am done when: The system updates the book status and increases available quantity.

US-15 – Automatic Fine Calculation

Story ID: US-15
Story Title: Calculate Overdue Fines
Importance: High
Estimate: 3 hours

User Story:
As a: System
I want: to calculate overdue fines automatically
So that: penalties are applied correctly

Acceptance Criteria:
And I know I am done when: The system calculates fines based on overdue days and updates the student's account balance.

US-16 – Pay Fine Online

Story ID: US-16
Story Title: Pay Fine Online
Importance: High
Estimate: 4 hours

User Story:
As a: Student

I want: to pay fines online
So that: my account remains active

Acceptance Criteria:
And I know I am done when: Payment is processed successfully and fine balance updates accordingly.

US-17 – Verify Fine Payment

Story ID: US-17
Story Title: Verify Fine Payment
Importance: Medium
Estimate: 2 hours

User Story:
As a: Librarian
I want: to verify fine payments
So that: records remain accurate

Acceptance Criteria:
And I know I am done when: Payment records are visible and marked as verified.

Reservation System

US-18 – Reserve Book

Story ID: US-18
Story Title: Reserve Unavailable Book
Importance: Medium
Estimate: 3 hours

User Story:
As a: Student
I want: to reserve unavailable books
So that: I can receive them when returned

Acceptance Criteria:
And I know I am done when: A reservation is successfully placed and the student is added to the waiting list.

US-19 – Manage Reservations

Story ID: US-19
Story Title: Manage Reservations
Importance: Medium
Estimate: 3 hours

User Story:
As a: Librarian
I want: to manage reservations
So that: books are allocated properly

Acceptance Criteria:
And I know I am done when: The librarian can view reservation queues and assign books to the next student.

US-20 – Generate Reports

Story ID: US-20
Story Title: Generate System Reports
Importance: High
Estimate: 4 hours

User Story:
As an: Admin
I want: to generate system reports
So that: I can monitor performance, usage, and fine collection

Acceptance Criteria:
And I know I am done when: The system generates downloadable reports showing book statistics, borrowing activity, and fine collection details.

6. Structured Specifications (Sample Core Stories)

Structured Specification – US11 (Request Book)

Preconditions:

- Student account is approved.
- Book exists in catalogue.

Main Flow:

1. Student logs into system.
2. Searches for desired book.
3. Clicks “Request Book”.
4. System records request as pending.
5. Librarian reviews request.
6. Librarian approves request.
7. System records issue date and due date.
8. Book quantity decreases.

Postconditions:

- Book marked as issued.

Alternate Flow:

- If book unavailable → student can reserve.

Structured Specification – US14 (Return Book)

Preconditions:

- Book must be issued.

Main Flow:

1. Librarian selects return option.
2. Enters book ID.
3. System calculates delay (if any).
4. Fine generated automatically.
5. Book quantity updated.

Postconditions:

- Book marked as returned.

Structured Specification – US15 (Fine Calculation)

Preconditions:

- Book returned after due date.

Main Flow:

1. System calculates overdue days.
2. Fine per day applied.
3. Fine recorded in student account.

Postconditions:

- Fine visible to student and librarian.

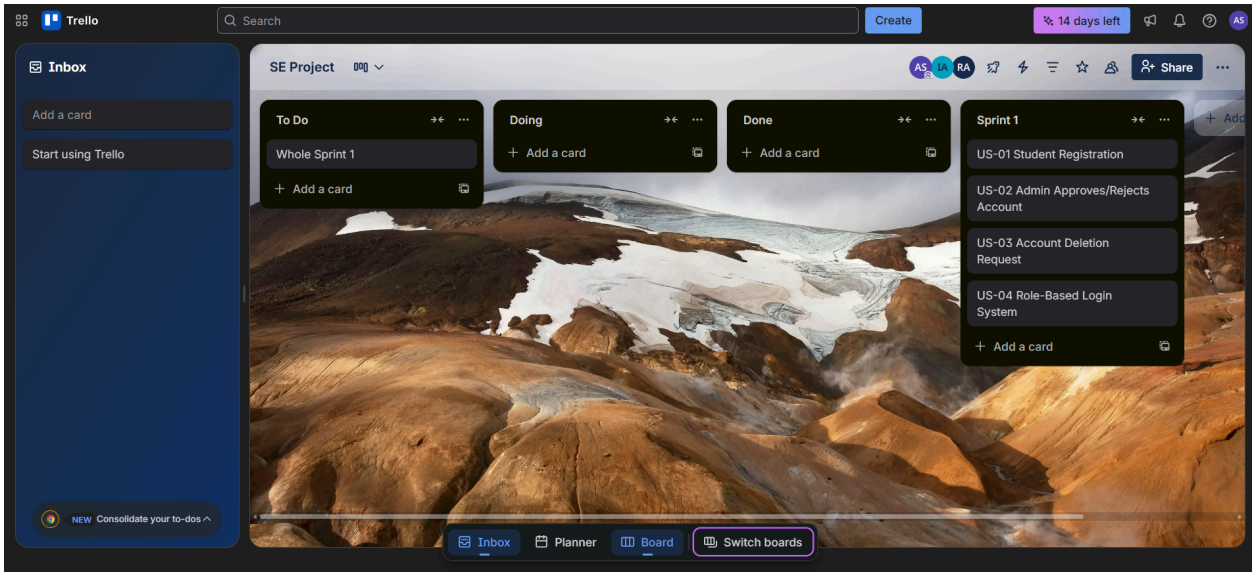
7. Scrum Board

Sprint 1 focuses on:

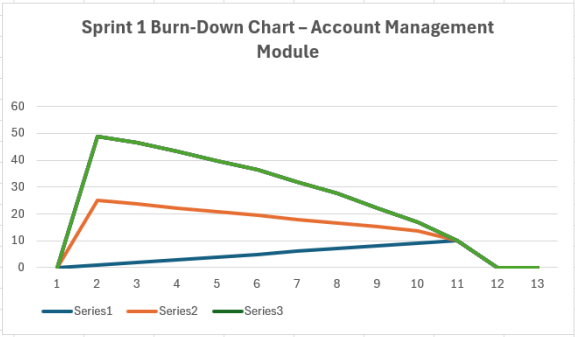
- Book Catalogue Module
- Basic Account Management
- Book Request Workflow

Trello Board Columns:

- To Do
- Doing
- Done
- Current Sprint



Day	Planned Tasks Remaining	Actual Tasks Remaining
1	24	24
2	21.6	23
3	19.2	21
4	16.8	19
5	14.4	17
6	12	14
7	9.6	11
8	7.2	7
9	4.8	3
10	0	0



8. Non-Functional Requirements (NFR)

Performance:

- System should respond within 2 seconds for search operations.

Security:

- Only authorized users can access role-specific features.
- Passwords must be securely stored.

Usability:

- Interface should be simple and user-friendly.

Reliability:

- System should prevent duplicate issuing of books.
- System should ensure data integrity.

Scalability:

- System should support addition of future modules.

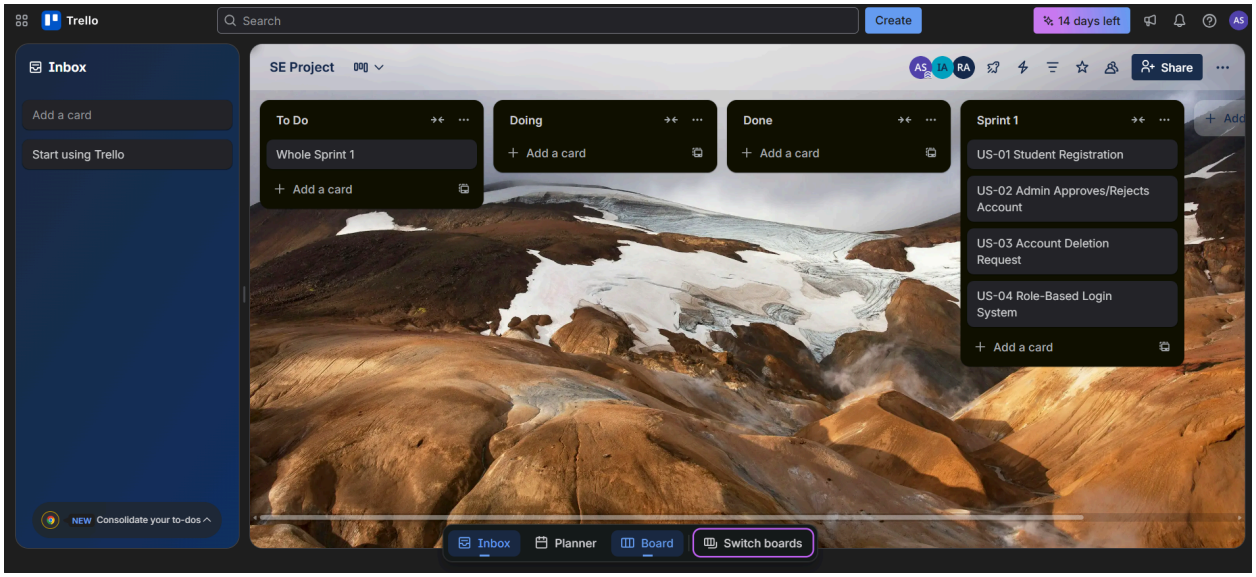
Maintainability:

- Code must follow object-oriented principles.

Availability:

- System should be operational during library working hours.

9. Iteration 1 Implementation Screenshots



10. Work Division

👤 **Name: Ali Bin Salman**

Role(s): Project Manager, Requirement Analyst, Developer

Responsibilities:

- Led overall project planning and coordination.
- Managed timeline, sprint goals, and ensured deliverables were completed on schedule.
- Conducted requirement gathering and analysis.
- Prepared user stories and structured specifications.
- Contributed to the implementation of the Account Management module.
- Assisted in backend logic for student registration and admin approval workflows.
- Participated in sprint planning meetings and decision-making processes.

👤 **Name: Muhammad Rohaan**

Role(s): Developer, Scrum Master, Architect

Responsibilities:

- Designed the overall system architecture.
- Designed database structure and defined relationships between entities.
- Ensured proper role-based access control implementation.
- Implemented core backend functionalities.
- Managed and maintained the Trello Scrum board throughout the sprint.
- Conducted daily stand-ups and monitored sprint progress as Scrum Master.
- Ensured Scrum practices were followed correctly.

👤 **Name: Muhammad Atyab**

Role(s): Tester, UI Designer, Developer

Responsibilities:

- Designed user interface layouts for registration, login, and admin dashboard.
- Implemented frontend components of the Account Management module.
- Performed functional testing of implemented features.
- Identified and reported bugs during sprint execution.
- Verified role-based login and account approval workflows.
- Assisted in debugging and refining system features.
- Contributed to documentation and screenshot preparation.

Shared Responsibilities (All Members)

- Requirement analysis and discussion.
- Sprint planning and task breakdown.
- Participation in Scrum meetings.
- Reviewing completed features before marking tasks as done.
- Contributing to documentation and report preparation.